Coalitional games

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Game Theory and Mechanism Design - Lecture 11

(This lecture heavily relies on Narahari (2014) Game Theory and Mechanism Design – chap. 27)

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

- Use the concept of **coalition**
- Define coalitional games **with transferable utility**
- Describe and discuss **properties** and **special structures** of coalitional games
- Give **examples** of coalitional games

Outline

1. Sharing the inheritance
2. Coalitional game with transferable utility
3. Imputation and domination
4. Monotonicity, super-additivity and convexity
5. Essential games

1. **Sharing the inheritance**

The set-up

One can find many stories about how inheritance may lead to some "intense" games and negotiations...

Imagine that

- we have 3 players $N = \{1, 2, 3\}$
- an inheritance of $L = 600$ (k£) to share among them
- The pay-offs of the players are u_i $(i = 1, 2, 3)$
- the person leaving the inheritance has placed different conditions in their will...
- these conditions lead to different versions of the ultimatum game

[Note: Nash's bargaining theory is great for the case of 2 players... but, will that work well for more than 2 players?]

First case: all aligned

The condition formulated by the person leaving the inheritance is that the 3 players should agree on the allocation.

What would be the outcome?

- in principle, any allocation u_i $(i = 1, 2, 3)$ such that $\sum_i u_i = L$ is a Nash equilibrium
- we can see it as a bargaining problem
- Nash's bargaining solution is $u_i = \frac{L}{3}$ $(i = 1, 2, 3)$... so £200k each.

Unintended consequences

The person leaving the inheritance had the idea that players 1 and 2 are the responsible grown-ups, while player 3 may not necessarily be trusted. The condition is then that players 1 and 2 should agree on the allocation...

What would be the outcome?

- in principle (again), any allocation u_i $(i = 1, 2, 3)$ such that $\sum_i u_i = L$ is a Nash equilibrium
- and, on paper, it looks like the same bargaining problem.
- However, since player 3 is not involved, there is no reason to account for their opinion.
- (300,300,0) is then also a Nash equilibrium

[Note: it shows us that Nash's bargaining theory is only valid for the grand coalition (i.e., $\mathcal{C} = N$)]

Majority voting

The person leaving the inheritance had the idea that it is impossible to expect the 3 players to agree. So, the condition is that at least 2 players (any of them) should agree on the allocation...

What would be the outcome?

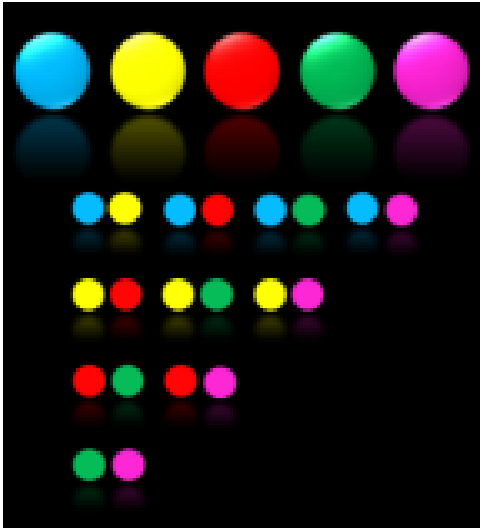
- Nash's bargaining solution is $u_i = \frac{L}{3}$ $(i = 1, 2, 3)$... so £200k each. However, there will be some weird dynamics developing
- For instance, players 1 and 2 may first (privately) agree on (300,300,0). For both of them, it is better than (200,200,200), so they are happy with it
- but then, player 3 may negotiate with player 2, and proposes (0,320,280). They are both better off than before, so they agree on it
- player 1 is back at it, contacts player 3, and proposes (200,0,400). They both are better off, hence they agree... etc. etc.

Do you see ways to get this under control?

2. **Coalitional game with transferable utility**

How many...

Given a set N of n players ($N = \{1, \ldots, n\}$), how many coalitions can be formed?



A coalition can be:

- any combination of 0 player
- any combination of 1 player
- any combination of 2 players
- ...
- any combination of n players

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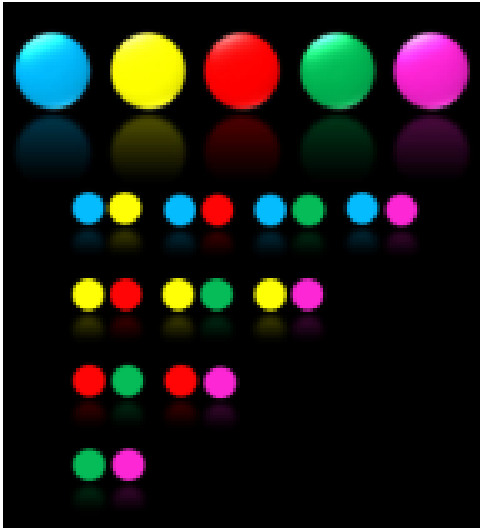

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Definition (power set)

Given a game with set of players $N = \{1, \ldots, n\}$, the *power set* 2^N is the set of all potential coalitions, i.e.

$$2^N = \{\emptyset, \{1\}, \ldots, \{1, \ldots, n\}\}\}$$

Its dimension is $|2^N| = 2^n$.

The number of coalitions with k players out of n players is given by $\binom{n}{k}$.

Characteristic function

Each coalition yields a certain value, captured by the characteristic function.

Definition (characteristic function)

Given a game with set of players $N = \{1, \ldots, n\}$, the *characteristic function* describes the joint pay-off (the so-called *value*) that can be obtained by each potential coalition $\mathcal{C} \subseteq N$ in the power set 2^N, i.e. $v : 2^N \to \mathbb{R}$. By convention, $v(\emptyset) = 0$.

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Example: coursework with rewards (possibly controversial)

- Consider a group of players $N = \{1, 2, 3\}$ attending the GTMD module
- The module leader tells them they are free to decide how to form groups.
- Their grade (0-100) will translate to a payment (in £) for each of the players
- The players can estimate the grade they believe can be reached depending on the groups...

Characteristic function:

$v(\emptyset) = 0$	$v(\{1\}) = 50$	$v(\{1,2\}) = 120$	$v(\{1,2,3\}) = 246$
	$v(\{2\}) = 55$	$v(\{2,3\}) = 146$	
	$v(\{3\}) = 65$	$v(\{1,3\}) = 144$	

(for instance, the pay-off of $\{1, 2, 3\}$ is due to a grade of 82, leading to 3 payments of £82, so 82x3=246)

Games with transferable utility

This is a game where we consider all potential coalitions that may form, and their reachable pay-offs

Definition (Coalitional game with transferable utility)

A *coalitional game with transferable utility*, denoted (N, v) is a game with a set of players $N = \{1, \ldots, n\}$ and a characteristic function $v : 2^N \to \mathbb{R}$, such that $v(\emptyset) = 0$.

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Example: inheritance game

In all cases we have $N = \{1, 2, 3\}$

All aligned:

$$v(\{\emptyset\}) = 0$$
$$v(\{1\}) = v(\{2\}) = v(\{3\}) = 0$$
$$v(\{1,2\}) = v(\{2,3\}) = v(\{1,3\}) = 0$$
$$v(\{1,2,3\}) = L$$

Majority voting:

$$v(\{\emptyset\}) = 0$$
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3. **Imputation and domination**

Coalitions have to make sense

Whatever coalition formation happens, the pay-offs are to be shared, eventually.

Definition (allocation)

Given a coalitional game with transferable utility (N, v), an *allocation* is the distribution of pay-off among all players, resulting from the coalition formation. That allocation is denoted $x = (x_1, \ldots, x_n)$.

Definition (imputation)

An *imputation* is an allocation that is both individually and collectively rational, i.e.

- $x_i \geq v(\{i\})$ (*individual rationality*)
- $\sum_i x_i = v(N)$ (*collective rationality*)

Collective rationality yields Pareto efficiency.

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Example: coursework with rewards

Remember the characteristic function:

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	$v(\{3\}) = 65$	$v(\{1,3\}) = 144$	

An example allocation is $x = (50, 82, 64)$ (e.g., based on player 1 delivering alone, while players 2 and 3 deliver a joint coursework report)... **Is that an imputation though?**

Domination

For some coalitions, certain imputations may look better than others.

Definition (domination of imputation)

Given a coalitional game with transferable utility (N, v), an imputation $x = (x_1, \ldots, x_n)$ dominates another imputation $x' = (x'_1, \ldots, x'_n)$ if there exists a coalition $\mathcal{C}$ such that

- $\sum_{i \in \mathcal{C}} x_i \leq v(\mathcal{C})$, and
- $x_i \geq x'_i, \ \forall i \in \mathcal{C}$. (with strict inequality for at least one of them)

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Here are 2 possible imputations

- (65,55,126) (based on 1 joint delivery, and "negotiated" sharing of the pot of money), and
- (50,55,141) (based on 1 joint delivery, and "negotiated" sharing of the pot of money).

Does one of these allocation dominate the other one?
Can you find other allocations that would dominate that one?

4. **Monotonicity, super-additivity and convexity**

Monotonicity

It always gets better when having larger coalitions, overall.

Definition (monotonic game)

A coalitional game with transferable utility (N, v) is said to be monotonic if, when enlarging a given coalition, the corresponding value of the coalition increases, i.e.

$$\mathcal{C} \subseteq \mathcal{D} \implies v(\mathcal{C}) \leq v(\mathcal{D}), \ \forall \mathcal{C} \subseteq \mathcal{D} \subseteq N$$

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Is the coursework with rewards game monotonic?

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Yes, it is!

Super-additivity

Even better, by having the value of a coalition being even greater than the sum of the values of its components.

Definition (super-additive game)

A coalitional game with transferable utility (N, v) is said to be super-additive if

$$v(\mathcal{C} \cup \mathcal{D}) \geq v(\mathcal{C}) + v(\mathcal{D}), \;\; \forall \mathcal{C}, \mathcal{D} \subseteq N, \; \mathcal{C} \cap \mathcal{D} = \emptyset$$

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Yes, it is!

Proposition (monotonic and super-additive games)

A super-additive game is necessarily monotonic. The opposite is not true.

Convexity

We can define games that are "even more"' than super-additive

Definition (convex game)

A coalitional game with transferable utility (N, v) is said to be convex if

$$v(\mathcal{C} \cup \mathcal{D}) + v(\mathcal{C} \cap \mathcal{D}) \geq v(\mathcal{C}) + v(\mathcal{D}), \quad \forall \mathcal{C}, \mathcal{D} \subseteq N$$

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Is this coursework with rewards game convex?

Remember the characteristic function:

$v(\emptyset) = 0$	$v(\{1\}) = 50$	$v(\{1,2\}) = 120$	$v(\{1,2,3\}) = 215$
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We need, in principle, to look at 3 cases (for the others, super-additivity is sufficient)... The first one would be

$v(\{1,2\} \cup \{1,3\}) + v(\{1,2\} \cap \{1,3\}) = v(\{1,2,3\}) + v(\{1\}) = 265$
This is while $v(\{1,2\}) + v(\{1,3\}) = 266$...
Hence, $v(\{1,2\} \cup \{1,3\}) + v(\{1,2\} \cap \{1,3\}) < v(\{1,2\}) + v(\{1,3\})$

No, it is not convex.

Proposition (convex and super-additive games)

A convex game is necessarily super-additive. The opposite is not true.

Convexity and snowball effect

The benefits from joining a coalition increase as the coalition gets larger... (snowball effect)

Proposition (convex games and individual additions)

A coalitional game with transferable utility (N, v) is convex if and only if, the benefit of adding an additional player to a given coalition gets higher for larger coalitions, i.e.

$$v(\mathcal{C} \cup \{i\}) - v(\mathcal{C}) \leq v(\mathcal{D} \cup \{i\}) - v(\mathcal{D}), \quad \forall \mathcal{C} \subseteq \mathcal{D} \subseteq N, \quad \forall i \in N \setminus \mathcal{D}$$

Proposition (convex games and group additions)

A coalitional game with transferable utility (N, v) is convex if and only if, the benefit of adding a group of players to a given coalition gets higher for larger coalitions, i.e.

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A convex version of the coursework with rewards game (characteristic function):

$v(\emptyset) = 0$	$v(\{1\}) = 30$	$v(\{1,2\}) = 85$	$v(\{1,2,3\}) = 200$
	$v(\{2\}) = 35$	$v(\{2,3\}) = 95$	
	$v(\{3\}) = 40$	$v(\{1,3\}) = 90$	

Indeed, we have

- $v(\{1,2\} \cup \{1,3\}) + v(\{1,2\} \cap \{1,3\}) = v(\{1,2,3\}) + v(\{1\}) \geq v(\{1,2\}) + v(\{1,3\})$
- $v(\{1,2\} \cup \{2,3\}) + v(\{1,2\} \cap \{2,3\}) = v(\{1,2,3\}) + v(\{2\}) \geq v(\{1,2\}) + v(\{2,3\})$
- $v(\{1,3\} \cup \{2,3\}) + v(\{1,3\} \cap \{2,3\}) = v(\{1,2,3\}) + v(\{3\}) \geq v(\{1,3\}) + v(\{2,3\})$

5. **Essential games**

Essential super-additive games

Inessential super-additive games are games of limited interest...

Definition (Essential super-additive games)

A super-additive game (N, v) is said to be *inessential* if $v(N) = \sum_{i \in N} v(\{i\})$.
Otherwise, it is called *essential*.

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Otherwise, it is called *essential*.

It is of limited interest, since for inessential games, there is a unique possible imputation

Proposition (Imputation in essential and inessential games)

An inessential game (N, v) has a unique imputation $(v(\{1\}), \ldots, v(\{n\}))$.
In contrast, an essential game can have an infinite number of imputations.

An inessential version of the coursework with rewards game (characteristic function):

$$v(\emptyset) = 0 \qquad \begin{array}{l} v(\{1\}) = 50 \\ v(\{2\}) = 55 \\ v(\{3\}) = 65 \end{array} \qquad \begin{array}{l} v(\{1,2\}) = 105 \\ v(\{2,3\}) = 120 \\ v(\{1,3\}) = 115 \end{array} \qquad v(\{1,2,3\}) = 170$$

Do you see any incentive to form coalitions?

Wrapping up

- Coalitional games focusing on **grouping of players** that may make sense
- Being part of a coalition yield a certain **value** (i.e., joint pay-off)
- All coalition values are distributed based on **pay-off allocations**
- An imputation is an allocation **that makes sense** (since individually and collectively rational)
- convex games $\subseteq$ super-additive games $\subseteq$ monotonic games

Further readings (/lecture notes for this topic):

Narahari Y (2014) *Game Theory and Mechanism Design* - chap. 27 (available on Blackboard Learn)

Thanks for your attention!